• Promote domestic manufacturing and export promotion in the Indian ESDM industry value-chain to build a globally-competitive ecosystem

• Increase exports to sixty per cent of domestic production by 2025

• Develop the manufacturing base for strategic electronics within the country

• Offer structured incentives to support the manufacturing of core electronic components

• Incentivise mega projects that are extremely high-tech and entail huge investments, such as semiconductor facilities for display fabrication

• Promote R&D and innovation across all emerging technology subsectors of electronics, such as 5G, the IoT, sensors, artificial intelligence (AI), machine learning, virtual reality (VR), drones, robotics, nano-based devices, etc

• Increase availability of skilled workforce, including re-skilling with appropriate incentives and support

• Develop such industries as fabless chip design, medical electronic devices, automotive electronics and power electronics for mobility and strategic electronics

• Promote development and acquisition of intellectual property rights or patents in the electronics sector to make them available to Indian entrepreneurs at a relatively low cost by creation of Special Sovereign Patent Fund (SPF)

Need of the hour

The Indian electronics manufacturing ecosystem is witnessing major changes from both regulatory and business points of view. However, these changes are not enough to materialise Make in India for the world.

According to a study by Deloitte, India is expected to be among the top five manufacturing nations by 2020. Interestingly, India's manufacturing labour cost in 2015 stood at US$ 1.72 per hour compared to US$ 37.96 per hour for the US. Even China's cost is almost double that of India.

The Indian government has set an aggressive target of increasing the manufacturing share to 25 per cent of GDP by 2025. However, till date, almost 75 per cent of the total electronics demand is still serviced through imports (Fig. 1). How should the Indian electronics manufacturing sector strategise to reach that place?

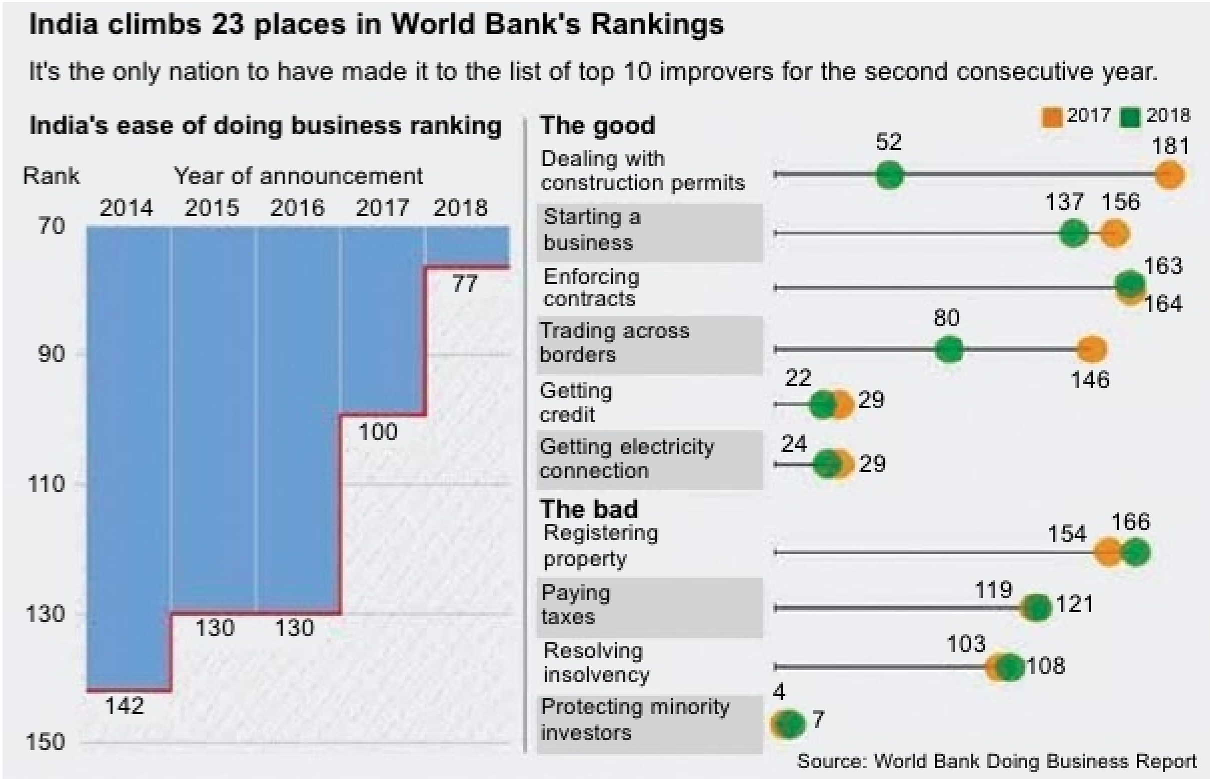

Fig. 2: India's improvement in World Bank's Ease of Doing Business ranking (Source: LiveMint)

For the Indian electronics manufacturing sector, it could also be the start of Industrial Revolution 4.0. The convergence between digital and manufacturing has already begun with wide adoption of the IoT, sensors, robotics, AI, etc. The industry also needs to reduce the cost of production through real-time automation and with the help of innovative technologies like 3D printing and so on.

The phased manufacturing programme (PMP) is also a good move to transform electronics manufacturing from semi-knocked-down (SKD) to completely-knocked-down (CKD) phase. But moving from SKD to CKD alone is not enough. The scale of manufacturing also needs to be increased, but that cannot happen by catering to the domestic market alone.

Companies need to be able to tap global markets, for which India needs a strong policy that provides incentives for the export of electronics goods. This will create the scale that will attract FDI into the country from global electronics component manufacturers and create jobs. This is what happened in auto manufacturing as well.

Incentives for the export of electronic goods will also be helpful in this aspect. Globally, manufacturing competitive countries like China, the UK, Australia, etc offer incentives, tax credits and grants to facilitate electronics manufacturing. The Indian government can adopt best practices from these countries to improve India's ranking on global manufacturing competitiveness (see table).

The ease of doing business could probably be the most important factor in establishing the country as a global hub for electronics manufacturing. In 2018, India moved to the 100th spot in World Bank's Ease of Doing Business global rankings due to sustained business reforms. However, global investors do not invest just on the basis of the overall ranking. They look at each parameter and every state's ranking on input parameters and do their own due diligence before investing.

The cost of logistics is also an important concern for investors. As per a World Bank report, cost of transporting one tonne of freight over one kilometre costs ₹ 2.28 by road, ₹ 1.41 by rail and ₹ 1.19 by water. Thus, efficient use of ports can help the country lower its logistics costs. This will, in turn, make domestic goods more competitive in global markets.

Other areas where the government could focus on is enforcing contracts. According to World Bank Ease of Doing Business survey, India now ranks 77; it jumped 23 positions in just one year.

If the Indian electronics sector wants to win this battle for global manufacturing supremacy, it has to focus on the above-mentioned areas. This will, in turn, drive innovation and cost competitiveness. EFY